

NanoGT Match Report: Fabry disease

Disease: Fabry disease (ORPHA:324)

Primary gene: GLA

Gene CDS: 681 bp

Inheritance: X-linked dominant

Target tissues: liver, kidney, heart, CNS

Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	RGX-121	AAV9	8.3/10	High	phase3
2	ABO-101	AAV9	7.7/10	High	phase1/2
3	SRP-9001	AAV9	7.5/10	High	approved
4	ST-920	AAV2/6	7.5/10	High	phase1/2
5	CPCB-RPE1	AAV8	6.9/10	Medium	phase2/3

Match #1: RGX-121

Precedent disease: Mucopolysaccharidosis type II

Vector: AAV9

Tissue target: CNS/liver

Composite score: 8.3 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	2.0	2.0
Tissue tropism	2.0	2.0
Protein class	0.5	2.0
Pathway similarity	2.0	2.0
Inheritance compatibility	1.0	1.0
Approval precedent	0.8	1.0

Rationale

- Gene CDS 681bp / cargo 4700bp (14% utilized)
- Vector tropism plus precedent target match: cns, liver
- Protein class mismatch
- Inheritance match (X-linked dominant <-> XL)
- Pathway match: lysosomal_storage
- Approval status: phase3

Match #2: ABO-101

Precedent disease: Mucopolysaccharidosis type IIIB

Vector: AAV9

Tissue target: CNS

Composite score: 7.7 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	2.0	2.0
Tissue tropism	2.0	2.0
Protein class	0.5	2.0
Pathway similarity	2.0	2.0
Inheritance compatibility	0.7	1.0
Approval precedent	0.5	1.0

Rationale

- Gene CDS 681bp / cargo 4700bp (14% utilized)
- Vector tropism plus precedent target match: cns
- Protein class mismatch
- LOF inheritance — compatible for gene replacement
- Pathway match: lysosomal_storage
- Approval status: phase1/2

Match #3: SRP-9001

Precedent disease: Duchenne muscular dystrophy

Vector: AAV9

Tissue target: muscle

Composite score: 7.5 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	2.0	2.0
Tissue tropism	1.5	2.0
Protein class	1.5	2.0
Pathway similarity	0.5	2.0
Inheritance compatibility	1.0	1.0
Approval precedent	1.0	1.0

Rationale

- Gene CDS 681bp / cargo 4700bp (14% utilized)
- Vector tropism overlap: cns, heart, liver
- Both membrane proteins
- Inheritance match (X-linked dominant <-> XL)

- Different pathway (lysosomal_storage vs myopathy)
- Approval status: approved

Match #4: ST-920

Precedent disease: Fabry disease

Vector: AAV2/6

Tissue target: liver

Composite score: 7.5 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	2.0	2.0
Tissue tropism	1.5	2.0
Protein class	0.5	2.0
Pathway similarity	2.0	2.0
Inheritance compatibility	1.0	1.0
Approval precedent	0.5	1.0

Rationale

- Gene CDS 681bp / cargo 4700bp (14% utilized)
- Precedent target match: liver
- Protein class mismatch
- Inheritance match (X-linked dominant <-> XL)
- Pathway match: lysosomal_storage
- Approval status: phase1/2

Match #5: CPCB-RPE1

Precedent disease: Achromatopsia

Vector: AAV8

Tissue target: retina/photoreceptor

Composite score: 6.9 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	2.0	2.0
Tissue tropism	1.5	2.0
Protein class	1.5	2.0
Pathway similarity	0.5	2.0
Inheritance compatibility	0.7	1.0
Approval precedent	0.7	1.0

Rationale

- Gene CDS 681bp / cargo 4700bp (14% utilized)

- Vector tropism overlap: cns, liver
- Both membrane proteins
- LOF inheritance — compatible for gene replacement
- Different pathway (lysosomal_storage vs retinal_phototransduction)
- Approval status: phase2/3